

HDOC Day-8 Report

Question 1: Number of Smaller Cubes

Problem Statement

Write a C program to find the number of smaller cubes formed by cutting a larger cube when both side lengths are given, assuming exact fitting.

Input

- Side of large cube
- Side of small cube

Output

- Number of smaller cubes

Formula

Number of cubes = $(\text{Large Side} / \text{Small Side})^3$

Test Cases

Test Case	Input(s)	Expected Output	Actual Output	Result
1	8,2	64	64	Pass
2	9,3	27	27	Pass
3	10,5	8	8	Pass

Algorithm

1. Start
2. Read large side
3. Read small side
4. Calculate cubes
5. Display result
6. Stop

Flowchart (Text)

Start → Input → Process → Display Output → Stop

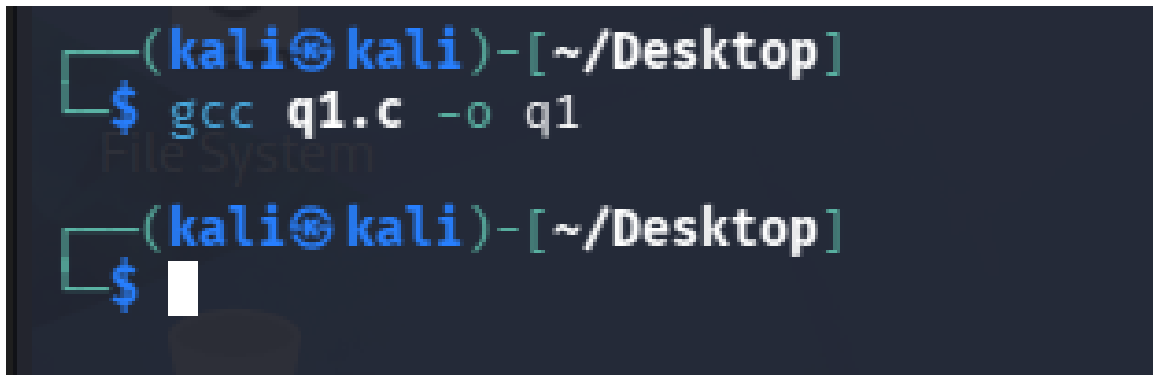
C Program

```
#include<stdio.h>
```

```
int main()
{
    int large, small, cubes;
    printf("Enter side of large cube: ");
    scanf("%d",&large);
```

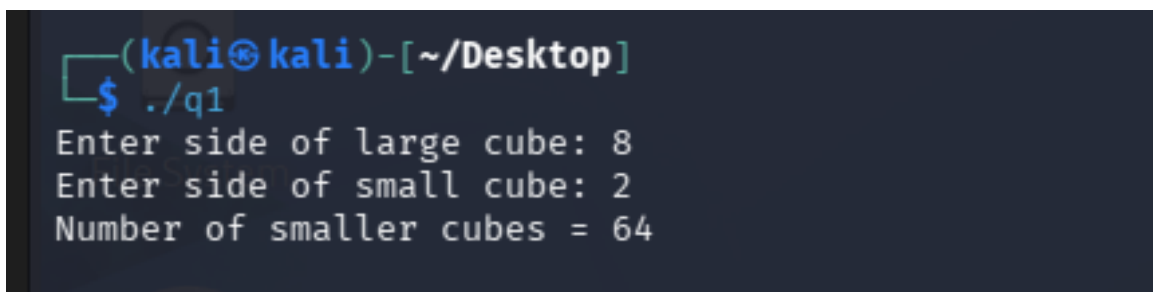
```
printf("Enter side of small cube: ");
scanf("%d",&small);
cubes=(large/small)*(large/small)*(large/small);
printf("Number of smaller cubes = %d",cubes);
return 0;
}
```

Compilation Screenshot



```
(kali@kali)-[~/Desktop]
$ gcc q1.c -o q1
(kali@kali)-[~/Desktop]
$
```

Execution Screenshot



```
(kali@kali)-[~/Desktop]
$ ./q1
Enter side of large cube: 8
Enter side of small cube: 2
Number of smaller cubes = 64
```

Question 2: Cost of Carpeting

Problem Statement

Write a C program to find the cost of carpeting a rectangular hall when length, breadth, and carpet rate per square meter are given.

Input

- Length
- Breadth
- Rate

Output

- Total carpeting cost

Formula

Area = Length $\times$ Breadth

Cost = Area $\times$ Rate

Test Cases

Test Case	Input(s)	Expected Output	Actual Output	Result
1	10,8,250	20000	20000	Pass
2	12,10,150	18000	18000	Pass
3	15,12,200	36000	36000	Pass

Algorithm

7. Start
8. Read length
9. Read breadth
10. Read rate
11. Calculate area
12. Calculate cost
13. Display cost
14. Stop

Flowchart (Text)

Start → Input → Process → Display Output → Stop

C Program

```
#include<stdio.h>
```

```
int main()
{
    float length,breadth,rate,area,cost;
    printf("Enter length: ");
    scanf("%f",&length);
    printf("Enter breadth: ");
    scanf("%f",&breadth);
    printf("Enter carpet rate per square meter: ");
    scanf("%f",&rate);
    area=length*breadth;
    cost=area*rate;
    printf("Total carpeting cost = %.2f",cost);
    return 0;
}
```

Compilation Screenshot

```
(kali㉿kali)-[~/Desktop]
$ gcc q2.c -o q2 -lm

(kali㉿kali)-[~/Desktop]
$
```

Execution Screenshot

```
(kali㉿kali)-[~/Desktop]
$ ./q2
Enter length: 10
Enter breadth: 8
Enter carpet rate per square meter: 250
Total carpeting cost = 20000.00

(kali㉿kali)-[~/Desktop]
```